

Quax

Game Instructions & User Manual

Quax is a two-player strategy board game played on an 11x11 grid. One player is human and the other is a computer-controlled bot. The two sides, **BLACK** and **WHITE**, take turns placing stones of their colour on empty tiles. There are two types of cells: **octagonal cells**, where stones are placed, and **rhombic (diamond) cells**, which act as diagonal connectors between adjacent octagons.

The winning conditions are simple: **BLACK** wins by creating a connected chain of black stones from the **top edge (row 11) to the bottom edge (row 1)**. **WHITE** wins by forming a connected chain of white stones from the **left edge (column A) to the right edge (column K)**.

Launch Instructions

The game was built and tested with **Java Development Kit (JDK) 25**. Ensure you have JDK 25 or a compatible version installed on your system

To launch the game:

1. Place Quax.jar in any folder on your computer.
2. Open a command prompt (Windows) or a terminal (macOS/Linux)
3. Navigate to the folder containing the JAR file.
4. Run the following command:

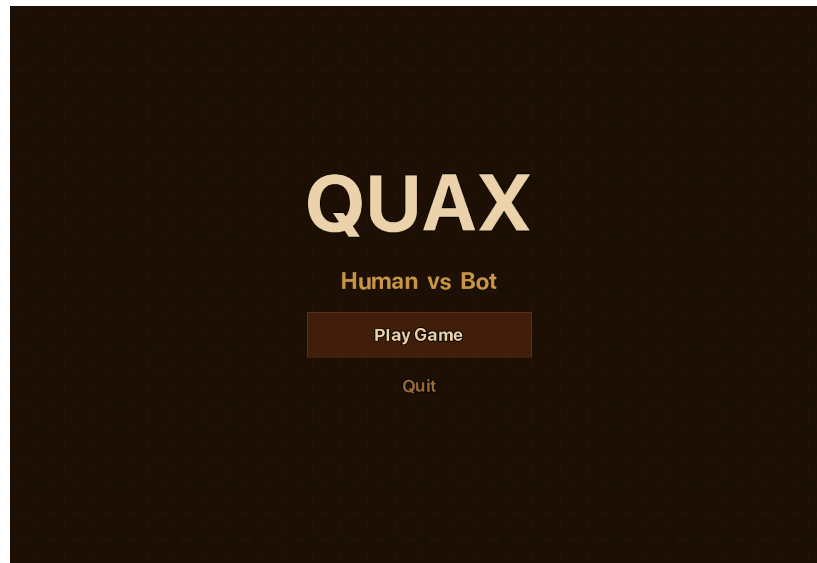
java -jar Quax.jar

5. The game window will open automatically after a moment.
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Welcome Screen

When the game launches you will see a dark welcome screen with the game title **QUAX**, a subtitle **Human vs Bot**, and two buttons:

- **Play Game** - starts a new game immediately.
 - **Quit** - exits the application
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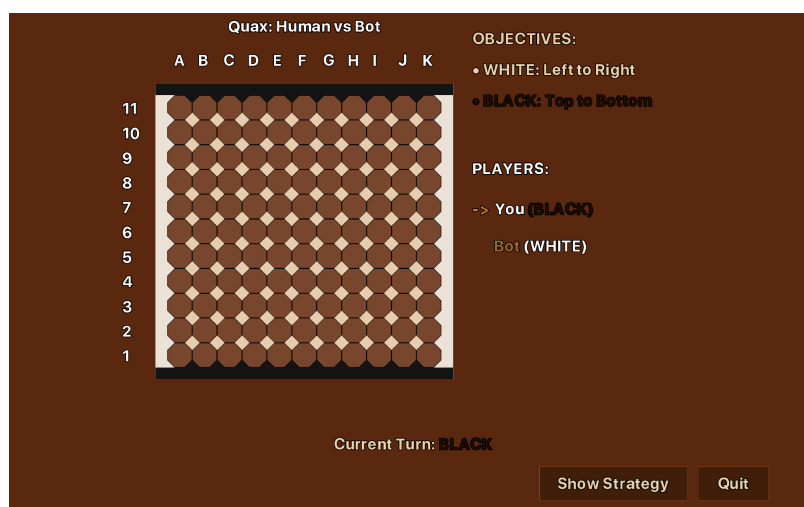


Welcome screen with Play Game and Quit buttons

Game Board Overview

After clicking **Play Game**, the main board appears with:

- An 11×11 grid of octagonal cells (initially empty).
- Rhombic (diamond) cells between diagonally adjacent octagons
- Row numbers **1-11** down the left side.
- Column letters **A-K** across the top.
- A **turn indicator** at the bottom centre, e.g., “Current Turn: BLACK”.
- A **player panel** on the right showing who is You and who is Bot, with their assigned colours and an arrow for the active player.
- An **objectives panel** on the right showing win conditions for each colour.
- A **Quit** button at the bottom-right corner.
- A **Show Strategy** button to the left of Quit.



Main game board at the start of a game

Making a Move

BLACK always moves first. At the start of each game, the bot is randomly assigned either black or white stones. If the bot is BLACK, it will place the opening stone automatically after a short pause. If the bot is WHITE, you play first as BLACK.

On your turn:

1. Hover over any empty cell - a yellow highlight shows which cell is selected.
2. Click the cell to place your stone. The cell immediately shows your colour.
3. The turn automatically passes to the bot, which will respond after a brief pause.

Note: You can place stones on both octagonal cells and rhombic cells. Rhombus tiles create diagonal connections between your stones.



Mid-game position with black and white stones placed and a cell (D6) being hovered over

Invalid Moves

If you click an occupied cell, a temporary message appears near the bottom of the screen: "**Invalid move. Select an empty cell.**" The message fades shortly after and your turn does not change.



Invalid move message displayed beneath the board

The Pie Rule

The pie rule makes the game fairer by allowing the second player to swap colours after the first stone is placed..

When does it appear?

Immediately after BLACK places the first stone, a brown button labelled “**Activate Pie Rule**” appears near the bottom-right corner of the screen. It is only visible during WHITE’s first turn.

For the human player (if you are WHITE on move 2):

- Click **Activate Pie Rule** before making your first move to swap colours.
- A confirmation banner appears for a few seconds confirming the swap.
- If you make a normal move instead, the button disappears permanently.

For the bot (if it is WHITE on move 2):

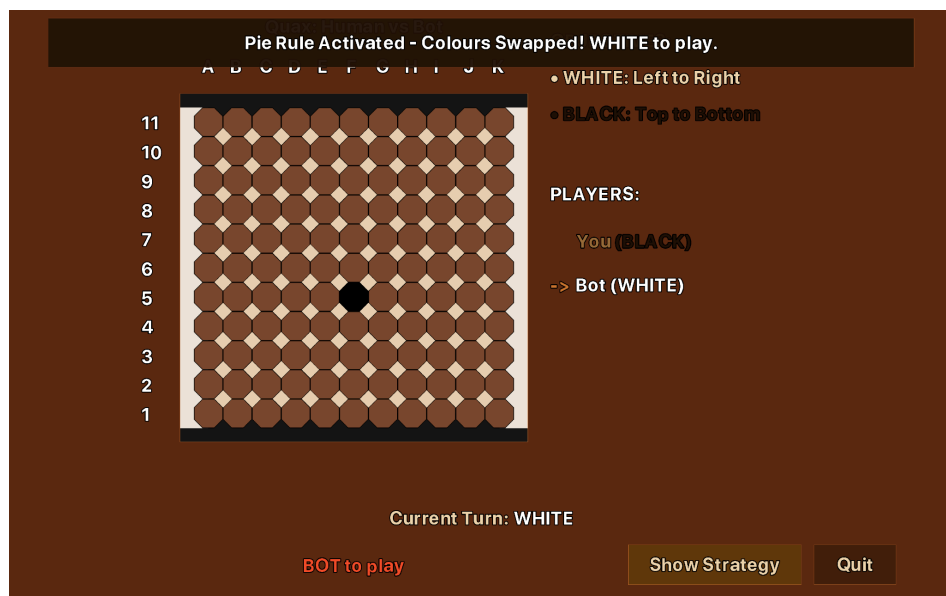
The bot has a **50/50 chance** of activating the pie rule. Sometimes it will swap colours, sometimes it will not and play a normal move. This keeps the game varied and unpredictable.

Can it be used later?

No. The pie rule is only available during WHITE’s first turn. After that, the button is gone for the rest of the game.



Activate Pie Rule button visible during WHITE's first turn



Confirmation banner after the pie rule is activated

Bot Behaviour

The bot uses a three-step strategy to choose every move:

1. **Win immediately** - if any move completes a winning chain, the bot plays it.
2. **Block the human** - if you are one move away from winning, the bot blocks that cell.
3. **Strategic positioning** - otherwise, the bot evaluates the entire board and plays the most contested cell - the position that is most valuable to both players simultaneously. This makes the bot disruptive throughout the game, not only near the end.

The bot considers both octagonal and rhombic cells equally. It will place a rhombus tile when doing so improves its strategic position. Opening moves are varied - the bot does not always start in the same spot.



"BOT to play" indicator shown at the bottom left while the bot calculates its next move

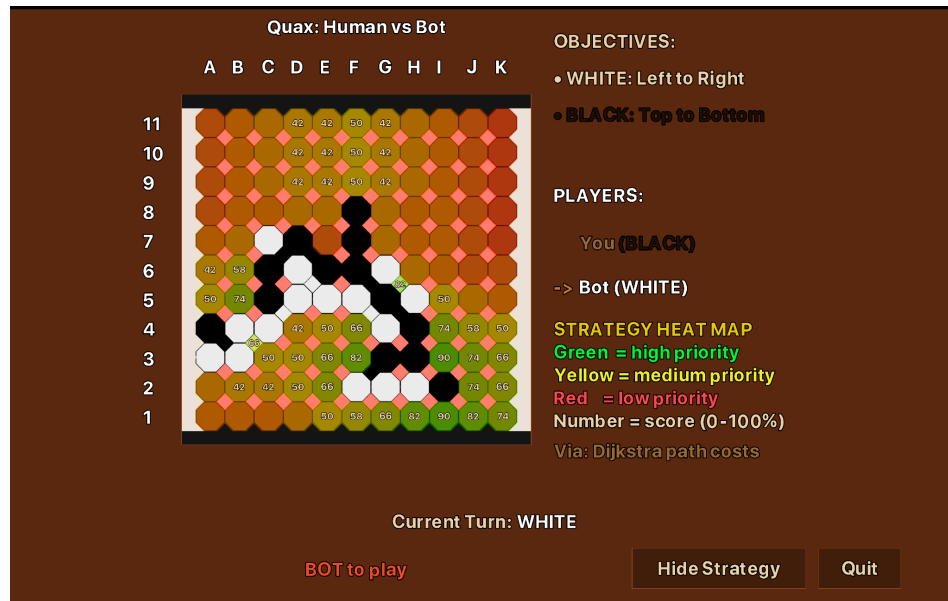
Strategy Visualisation

The game includes a heat map overlay that shows how the bot rates every available cell at the current moment.

How to enable it:

1. Click the **Show Strategy** button at the bottom of the screen
2. A colour overlay appears on every unoccupied cell:
 - **Green** = high priority - the bot wants this cell most
 - **Yellow** = medium priority
 - **Red** = low priority
3. Each coloured cell shows a **numeric score from 0 to 100%**.
4. A legend in the right panel explains the colour coding.
5. Click **Hide Strategy** to turn the overlay off.

The overlay updates after every move so it always reflects the current board state. Enabling or disabling the overlay has no effect on gameplay.



Strategy heat map enabled with colour-coded cell priorities

Winning the Game

The game continuously checks for a winning chain after every move.

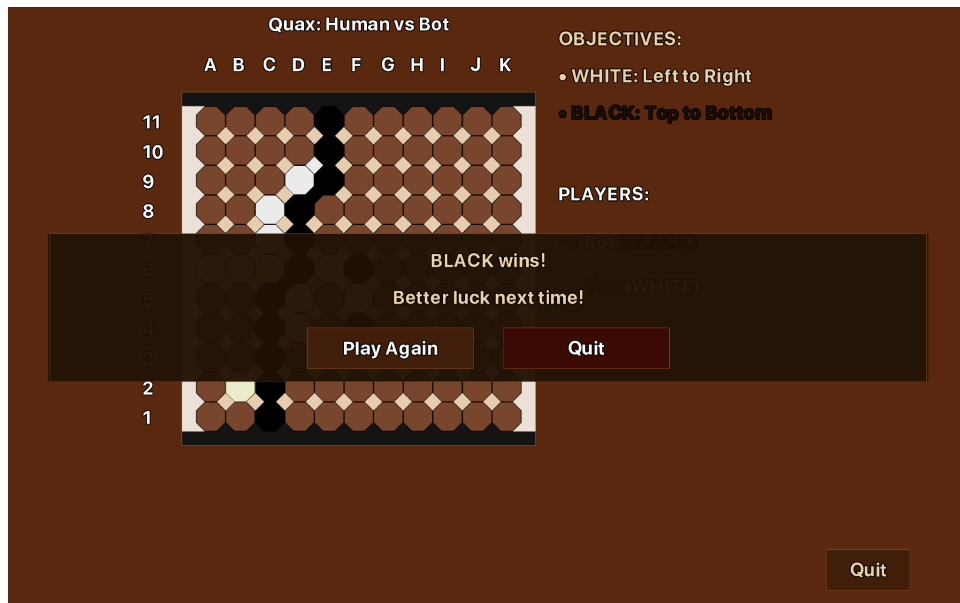
- **BLACK wins** by connecting **row 11 (top) to row 1 (bottom)** with a chain of black stones.
- **WHITE wins** by connecting **column A (left) to column K (right)** with a chain of white stones.

Two stones are connected when they:

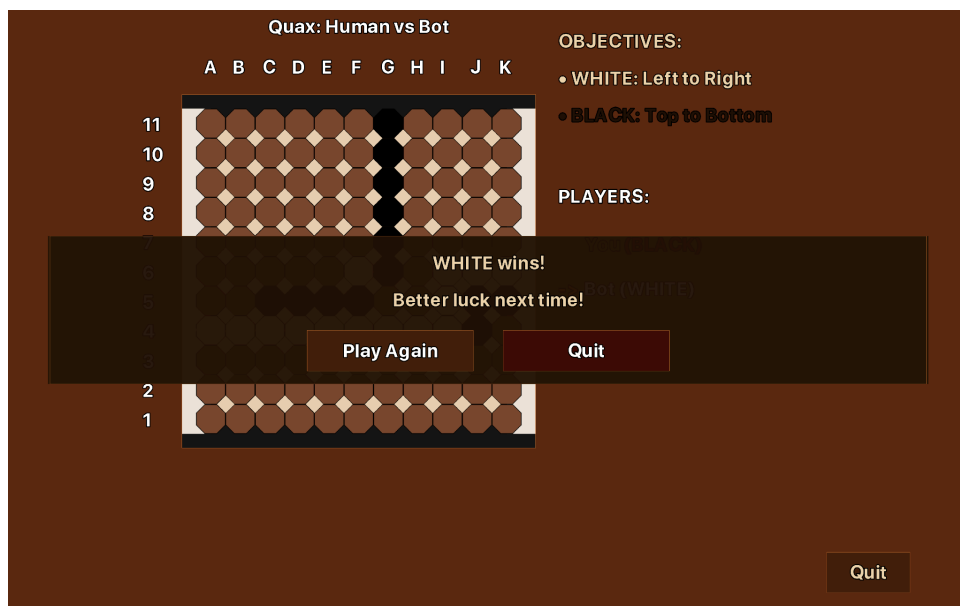
- Orthogonally adjacent (share a flat edge), or
- Are diagonally adjacent **and** a rhombus tile of the **same colour** sits between them

As soon as a winning chain is formed, the game immediately:

- Displays a banner: "**BLACK wins!**" or "**WHITE wins!**" in the winner's stone colour.
- Shows "**Congratulations!**" if you won, or "**Better luck next time!**" if the bot won.
- Disables all board interactions - no more stones or rhombus tiles can be placed.
- Shows two buttons: **Play Again** and **Quit**.



Win banner when BLACK wins



Win banner when WHITE wins

Playing Again

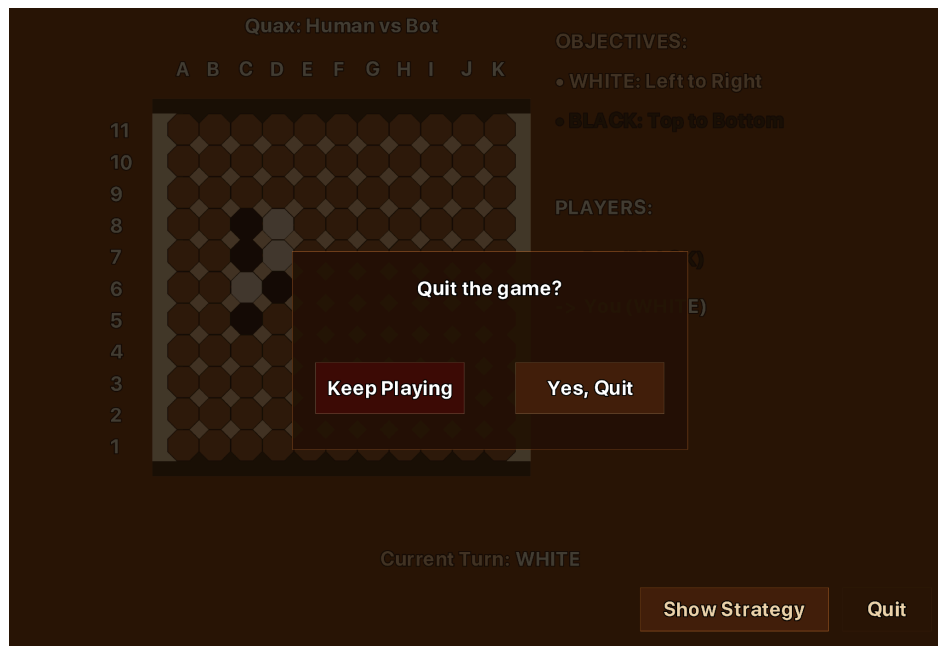
After a game ends you have two options:

- **Play Again** - resets the entire board and starts a fresh game immediately without closing the application. The bot is randomly reassigned a colour for the new game.
- **Quit** - exits the application.

Quitting the Game

You can quit at any time during a game:

1. Click the **Quit** button at the bottom-right corner
2. A confirmation dialog appears (only during the game):
 - **Keep Playing** - closes the dialog and resumes the game
 - **Yes, Quit** - closes the application



Quit confirmation dialog

Enjoy the game!